



# INTELLIGENT CANDIDATE RANKING SYSTEM

Ranking 100,000 candidates for a Senior AI Engineer role

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# OBJECTIVE OF THE PROJECT

## Problem

- Recruiters go through 100K profiles manually
- Keyword filters miss qualified candidates
- Can't detect fake/honeypot profiles

## Our Solution

- Multi-signal scoring across 5 dimensions
- Honeypot + keyword stuffer detection
- 23 behavioral signals from redrob platform

## Output

- Top 100 candidates ranked with scores
- Human-readable reasoning per candidate
- Runs in under 60 seconds on CPU

# PROBLEM AND PRIOR RESEARCH

## PROBLEM

- 100,000 candidates — impossible to review manually
- Keyword matching misses hidden gems
- No way to detect fake/stuffed profiles
- Behavioral signals completely ignored

## PRIOR RESEARCH

- Basic cosine similarity → misses context
- Pure embedding search → ignores experience
- TF-IDF → no fraud detection
- LangChain wrappers → too slow, needs API

# ADDRESSING RESEARCH GAPS



## Skill Depth

- Skills weighted by proficiency AND duration
- Expert + 1 month < Intermediate + 2 years



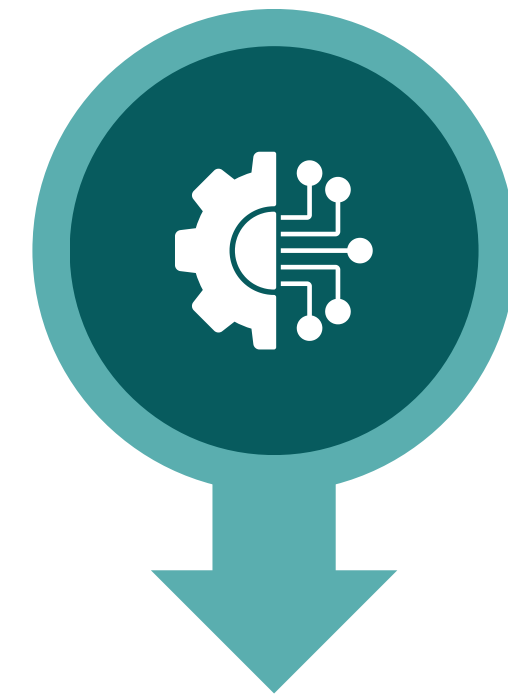
## Fraud Detection

- 6 honeypot detection rules
- Keyword stuffer penalty up to -0.40



## Behavioral Signals

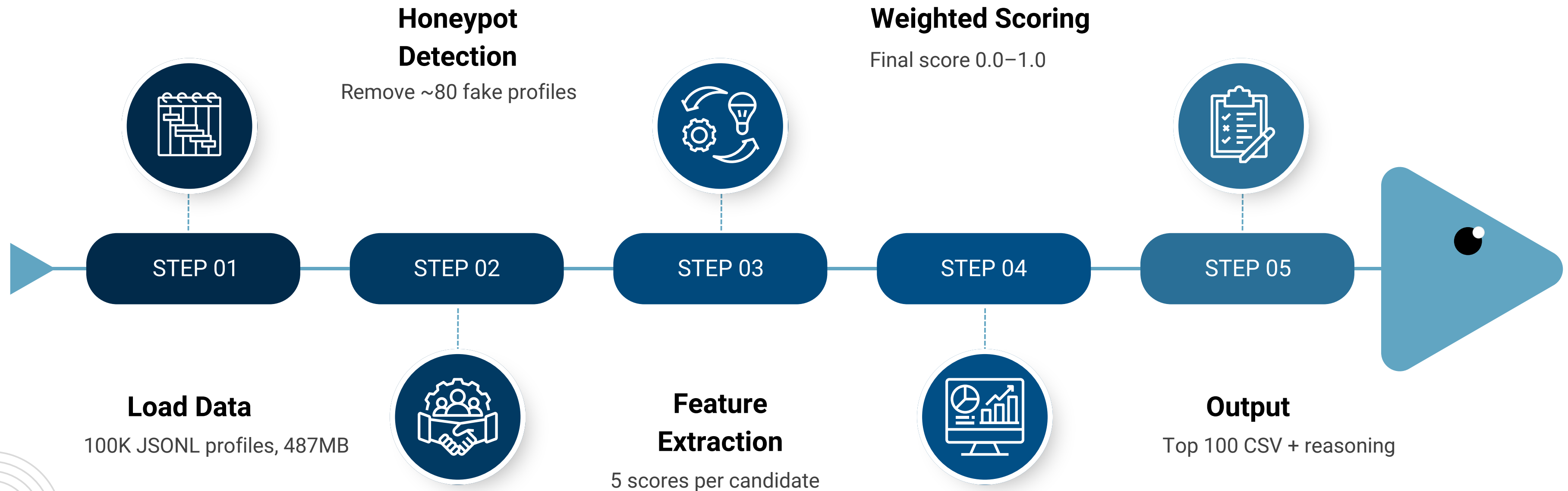
- 23 redrob signals used
- Recruiter response rate, recency, GitHub



## Explainability

- Every candidate gets real reasoning
- Not "good candidate" — actual data points

# PROJECT METHODOLOGY







# **DATASET** **DESCRIPTION**

## **Dataset Used:**

- 100,000 candidate profiles in JSONL format
- Each profile has: skills, career history, education, redrob signals

## **Profile Structure:**

- Skills — name, proficiency, duration\_months, endorsements
- Career History — company, title, description (free text)
- Education — institution, tier (Tier 1 to Tier 4), field
- Redrob Signals — 23 behavioral fields

## **Key Numbers:**

- 100,000 total candidates
- ~80 honeypots detected and removed
- 99,920 clean candidates scored
- Top 100 selected for submission